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Instructions From an Active Robotic Hand Increase Body Ownership and Task Clarity

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***Abstract*—This study introduces a novel approach to home rehabilitation exercise instructions, utilizing a 3D-printed robotic hand that can mimic the patient's hand movements and follow pre-programmed instructions. This robotic hand, made from inexpensive 3D printed components and hobbyist equipment, is controlled by open source computer vision tracking.**

Two within-subject crossover studies were conducted with 30 healthy participants. The first study assessed the participants' sense of embodiment, while the second compared their ability to remember movement exercises when instructed by the robotic hand versus video clips.

The results showed that the participants experienced a higher sense of embodiment with the robotic hand in its active mode. Furthermore, they demonstrated significantly better short-term memory retention of medium and difficult finger movement sequences when instructed by the robotic hand compared to video instruction ($F_{1,56}$ = 4.047, p $<$ .05) and ($F_{1,56}$ = 19.463, p $<$ .05). These findings suggest that this innovative method could improve remote therapy and strengthen bodily ownership, which is essential in stroke rehabilitation.

***Index Terms*—Body Ownership Illusion, Embodiment, Human-Robot Interaction, Rubber Hand Illusion, Stroke Rehabilitation**

I. Introduction

Rehabilitating finger and grip movements at home is relevant for several patient groups, for example, stroke survivors. Stroke can cause paralysis or reduced muscle control in the hand, which requires rehabilitation to regain strength and coordination. According to the World Health Organization, approximately 15 million people suffer strokes annually worldwide [1]. A significant proportion of stroke survivors experience hand weakness or motor deficits, indicating a substantial number of patients who need hand rehabilitation services. Furthermore, stroke is one of the leading causes of long-term severe disabilities, underscoring the need for effective rehabilitation strategies. Despite the availability of therapeutic interventions that can ameliorate post-stroke disabilities, a significant proportion of survivors continue to face chronic physical or cognitive impairments. For example, approximately 65% of the survivors struggle to reintegrate their affected hand into daily activities after completing conventional therapy [2]. This scenario is set against a backdrop of increasing costs, with stroke-related expenses in the European Union expected to grow significantly from an estimated 45 billion euros in 2017 [3].

In addition, hand rehabilitation is commonly needed after surgeries. Patients recovering from hand injuries, such as fractures, dislocations, or severe sprains, require rehabilitation to regain strength and dexterity. As the aging population increases, so does the prevalence of age-related hand problems. Conditions such as osteoarthritis and rheumatoid arthritis, both known to impact hand movement, are relatively common; consequently, a significant number of older adults are likely to seek hand rehabilitation services. Parkinson's disease can also affect hand function, necessitating ongoing rehabilitation. Finally, age-related weakness and loss of dexterity in the hands are common among the elderly, making exercises crucial to maintaining joint mobility and reducing pain.

The primary objective of this paper is to support the rehabilitation of late-stage stroke, carried out within the home environment. This phase lacks a clear end point and may last for years, relying heavily on the motivation, effort, and set goals of the survivor [4]. We relate to mirror therapy, since our robotic hand acts as a direct alternative to mirror therapy with the added benefit of being able to personalize the rehabilitation of the patient by having them interact with their therapist remotely.

The rubber hand illusion (RHI) [5] is an important method in investigating body ownership and agency and plays a crucial role in the effectiveness of mirror therapy. In this experiment, we successfully elicited the RHI using a robotic hand, demonstrating a key component to leverage mirror therapy techniques.

Rehabilitation robotics offers promising prospects for improving therapeutic interventions and addressing the challenges of teletherapy. By allowing therapists to remotely interact with patients through the robotic hand, our goal is to improve the personalization and effectiveness of home rehabilitation. The development of a 3D robotic hand instructor that is controlled by stroke patients and accessible remotely to therapists exemplifies this approach, aiming to surpass the limitations of conventional therapy [12] [13].

Our study investigates the potential for a 3D printed robotic hand to induce a sense of embodiment in participants, similar to the illusion of a rubber hand [5], and explores its effects

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Fig. 1. The Robotic Hand connected to an Arduino Uno, alongside an Intel RealSense D435 Stereovision Camera.

on performance in memorizing task instructions. In this way, we focus on developing user-friendly teletherapy support systems that provide clear instructions, ensuring accessibility for individuals with limited digital literacy.

II. Related work

The rubber hand illusion demonstrates how sensory information can manipulate the feeling of body ownership, where participants come to accept a rubber hand as part of their own body through a combination of visual, tactile, and proprioceptive stimuli. This illusion is induced by simultaneously stroking both the participant's hidden real hand and a visible rubber hand, leading to an altered perception of self-ownership over the artificial limb. Investigating the limits of self-perception via the rubber hand illusion and body ownership illusions allows researchers to uncover significant insights into the brain's adaptable representation of self. Research has extended the RHI to virtual bodies [6] and explored the use of robotic hands to induce similar illusions [7] [8]. The relationship between ownership and agency in these illusions has been investigated [9], with findings suggesting that these are distinct but interacting cognitive processes. Furthermore, the embodiment of robotic or virtual hands has been explored in the field of psychology and robotics [10] [11]. These studies highlight the potential for technology to manipulate body perception and create novel rehabilitation strategies.

Mirror therapy is a therapeutic technique used to alleviate sensory and motor deficits, particularly in the affected limb, by employing the principle of visual illusion. A mirror is strategically placed to reflect the image of the unaffected limb, creating the illusion that the affected limb is moving without difficulty [14]. Mirror therapy capitalizes on neuroplasticity in the brain, promoting the re-wiring of neural pathways associated with motor function and sensory perception [15]. By visualizing the affected limb in a normal state, the brain is encouraged to restore connections and improve motor control, which has shown promise in improving motor function of the limb, reducing pain perception, and improving overall recovery of the upper limb [16] [17]. However, therapy success depends on the patient's ability to perform exercises correctly, which, in some cases, requires guidance and practice under the supervision of a skilled healthcare professional [18]. This required guidance ensures that patients engage in mirror therapy with precision, optimizing its potential benefits, while mitigating the risk of incorrect execution. Furthermore, the applicability of mirror therapy may vary from person to person. Some people do not respond as well as others to mirror therapy [19]. Consequently, the utility and generalizability of mirror therapy in the context of hand stroke rehabilitation remain subject to review and ongoing investigation [20].

Teletherapy serves as a powerful conduit to transcend geographical barriers that have historically restricted access to expert stroke rehabilitation services [21]. The ability to receive therapy remotely ensures that people residing in remote or underserved areas, often far away from specialized rehabilitation centers, can access the specialized care they require. This geographical inclusivity addresses a disparity in healthcare access, promoting equity and ensuring that stroke survivors from various locations can benefit from the latest advances in upper limb stroke rehabilitation [22]. In addition, teletherapy offers the flexibility of scheduling sessions at times convenient to the patient, further enhancing adherence and overall therapeutic results. Despite its considerable advantages, teletherapy faces a series of challenges in the context of upper limb stroke rehabilitation, the most notable being the limited physical interaction it provides. The absence of physical interaction in teletherapy poses a fundamental limitation, particularly in cases where personalized tactile interventions are crucial to rehabilitation success [23].

The effectiveness of teletherapy is intricately intertwined with technological prerequisites. Patients must have access to suitable equipment and possess a degree of digital literacy to participate effectively in remote therapy sessions. Although advancements in technology have made these prerequisites increasingly accessible, disparities persist, potentially hindering certain individuals' participation. The digital divide, compounded by socioeconomic disparities, can exacerbate healthcare inequalities, raising relevant ethical concerns [24].

III. Material and Methods

The robotic hand is constructed using open source 3D prints from the InMoov project [25] and Parallax Inc 4 servomotors.

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The robotic hand is operated by implementing computer vision, using the OpenCV library (Version 4.8.1), to detect hand movement, which is then translated into binary finger movement (i.e., open or closed) in the robotic hand. The script, developed in Python, utilizes an Intel D435 camera along with MediaPipe for hand tracking and palm detection, selected for its superior image quality and speed, making it well-suited for tracking rapidly moving fingers. It detects 21 hand landmarks to identify open or closed fingers based on their positions. Finger counting is performed using a skeletal mesh model, and the status of fingers is determined by comparing the positions of the fingertips and phalanges. The script then sends a string to an Arduino script to represent the hand's state, with each digit in the string representing a finger's status (open or closed).

The script imports libraries for computer vision and hand tracking, including 'cv2' for image processing, and custom modules from 'cvzone' (Version 1.6.1) for hand detection and tracking. It initializes video capture, sets parameters for hand detection, and uses a loop to capture frames, detect hands, and count open fingers. The hand detection results are used to control an Arduino Uno via serial communication, displaying the video feed in real-time with overlaid hand landmarks and the detected hand contours.

The Arduino script controls the servomotors based on the data received from the hand-tracking script. It reads serial data, identifies commands for finger positions, and adjusts the servomotors accordingly. The setup includes an Arduino Uno, Parallax servomotors, and an Adafruit servo shield, chosen for their cost-effectiveness, quality, and capability to handle multiple servomotors with an external power supply. The system translates digital instructions into motion, controlling servomotors to replicate hand gestures. The total cost of the components amounts to approximately €525, with the Intel D435 stereo camera being the most financially significant component at around €390. However, a basic webcam can be implemented as a viable substitute, reducing the total cost to €150. Although this cost reduction slightly compromises image quality, it still enables satisfactory finger tracking and movement detection, making it a practical and cost-effective alternative. This approach is significantly more affordable than virtual reality mirror therapy sets, which range from $252 [26] to considerably higher prices depending on technological complexity [27]. In particular, the system achieves a minimal latency of 90 milliseconds, significantly below the 300-millisecond threshold beyond which visuomotor delays become perceptible to users [28].

The research comprises two controlled within-subject crossover studies aimed at assessing the participants' sense of embodiment and their memorization performance when interacting with the robotic hand. All participants were recorded as right-hand dominant. Before starting the experiments, participants were asked about their gender and age group, 22 participants being male (73%) and the other 8 female (27%). Most of the participants belonged to the 18-24 age group (53%), the rest was made up of the 25-34 age group (37%), the 35-44 age group (7%) and the 45-54 age group (3%). All participants were university students, except for one. Before participation, all participants filled out a signed consent form. This research work was exempted from ethical approval by the Danish Research Ethics Committee because data was collected solely for technical purposes and were not intended for use in a clinical setting.

The methodology involved dividing participants into two groups, each undergoing a test and control study in both experiments, with the order of participation reversed between the groups. The two experiments were conducted successively, starting with the first experiment.

Experiment 1 sought to determine whether participants experienced a sense of embodiment with the robotic hand. Participants were seated with their hands on a table in front of them, but unlike the usual procedure their hands were facing upwards, not downwards. A divider and a long-sleeved shirt worn backward were used to obscure their right hand and replace its visual position with the robotic hand (Figure 2). This setup was aimed at minimizing distractions and maintaining the illusion. An Intel RealSense D435 camera captured the movements of the participants' hands, allowing them to control the robotic hand. The experiment consisted of one control phase and one test phase, each lasting 5 minutes. In the control phase, both the participant's right hand and the robotic hand were simultaneously stroked with a paintbrush to elicit the rubber hand illusion, and in the test phase, participants had full control over the robotic hand's movements. The order of the control and test phases was counterbalanced between the participants to minimize order effects. Half of the participants performed the control phase first, followed by the test phase, while the other half performed the test phase first, followed by the control phase. Both phases were followed by a questionnaire based on the questionnaire used by Botvinick and Cohen [5] to capture the participant's experience. Questionnaire responses were collected using a 7-point Likert scale, with answers numerically translated such that 7 represented 'strongly agree' and 0 represented 'strongly disagree'. A Wilcoxon Signed-Rank test was used to analyze the data from Experiment 1. In the control study, the robotic hand is referred to as ***passive***, since the participant is stimulated by touch administered by the researcher through paintbrushes. For the test study, the robotic hand is referred to as ***active***, since the participant is stimulating the robotic hand themselves by movement of the fingers of their right hand (Figure 2).

Experiment 2 focused on assessing whether interaction with the robotic hand improved participants' memorization task performance compared to observing a prerecorded video of a human hand. The setup included an iPad with video clips for the control study and the robotic hand for the test study, both displaying sequences of finger movements classified into easy, medium, and difficult instructions. The robotic hand was operated by the researcher. A task of easy complexity encompassed three to four distinct operations, while tasks of medium and difficult complexity comprised six and nine

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Fig. 2. Front view (left) and top view (right) of the setup of Experiment 1. Following the standard rubber hand illusion procedure, a divider hides their right hand, replacing it with the robotic hand.

operations, respectively. Participants were exposed to a total of 15 sequences across these categories and asked to remain still (no finger movements) during observation in both video and robotic hand conditions. They were then requested to replicate each sequence with their right hand after observing the demonstrated sequence. For each sequence in Experiment 2, a participant's score was calculated as the percentage of correctly replicated finger movements out of the total number of movements in that sequence. The sequence presentation order was randomized using a Latin Square design to avoid order effects. An ANOVA was used to analyze the data from Experiment 2.

IV. Results

Before the analysis of data from Experiment 1, participants were first screened for 'responders' and 'rejecters' based on the work of Kalckert and Ehrsson [29]. Rejectors are classified according to the criteria that they did not answer a single individual question on the positive spectrum of the Likert scale. Although some strong and mild responders were found, no rejecters were found in the data. All 30 participants were used in the analysis. The results indicated a significant sense of embodiment with the active robotic hand in three aspects: feeling of having an additional hand or arm (Passive: Mean = 3.5, SD = 1.77; Active: Mean = 4.5, SD = 1.7), sensation of the hand becoming 'mechanical' or 'artificial' (Passive: Mean = 3.6, SD = 1.65; Active: Mean = 4.7, SD = 1.7), and the resemblance of the robotic hand to the participant's own hand (Passive: Mean = 2.7, SD = 1.3; Active: Mean = 3.1, SD = 1.6). Furthermore, 20 of the 30 participants felt that the active robotic hand was more integrated into their body image compared to the passive robotic hand.

Data from Experiment 2 underwent variance analysis (ANOVA) to assess performance in a memorization task with either video instructions from the iPad or robotic hand guidance. No significant differences were observed in the easy tasks between mediums (iPad: 97.47%, Robotic Hand: 98.62%; $F_{1,56}$ = 1.373, ns). However, significant differences were observed in medium (iPad: 83.56%, Robotic Hand: 89.04%; $F_{1,56}$ = 4.047, p $<$ .05) and difficult tasks (iPad: 67.18%, Robotic Hand: 79.89%; $F_{1,56}$ = 19.463, p $<$ .05), indicating enhanced performance with the robotic hand. No significant training effect was found.

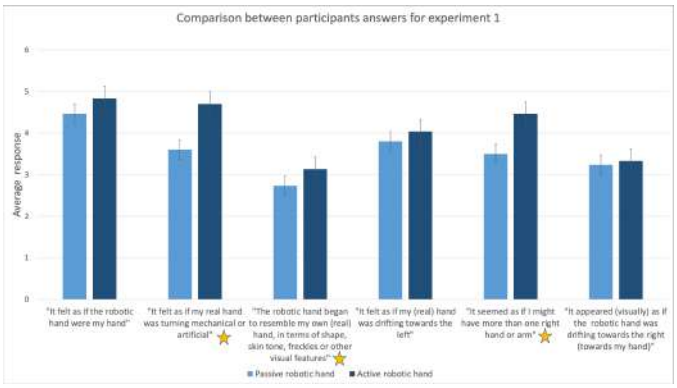

Fig. 3. Answers for 6 questions from the questionnaire proposed by Botvinick and Cohen [5]. A star marks results with a significant difference (p $<$ .05) between the passive (light blue) and active (dark blue) version of the robot hand.

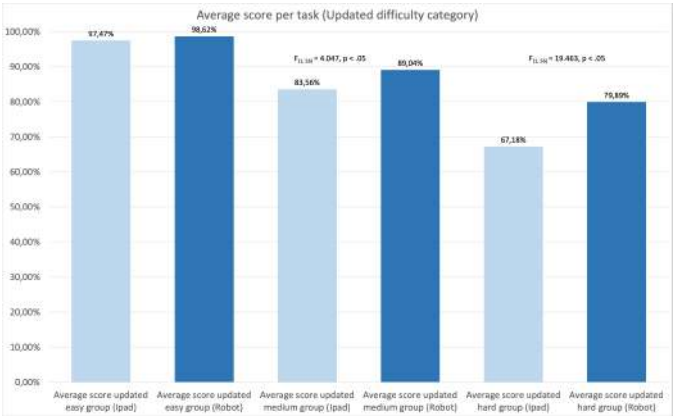

Fig. 4. Average correct replications of finger movement instructions per task (easy, medium, difficult) are shown. Light blue columns represent video clip instructions, and dark blue columns represent robotic hand instructions.

V. Discussion

Experiment 1's findings align with the foundational rubber hand illusion experiments by Botvinick and Cohen [5], demonstrating that the active robotic hand fosters a greater sense of embodiment and immersion compared to its passive counterpart. Participants reported a heightened sense of embodiment with the active robotic hand, in particular, noting its more convincing appearance and the sensation of having an additional limb. These results suggest that active interaction with the robotic hand not only reinforces the illusion of ownership, but also enhances the sensory integration necessary for a convincing experience of body ownership.

Qualitative feedback from participants further underscores the illusion's efficacy, with reports of the active hand feeling like a part of their body, indicating a strong sense of embodiment. However, discrepancies in timing and mechanical feedback occasionally disrupted this illusion, highlighting areas for improvement in the robotic hand's design.

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Arata et al. [8] also demonstrated the RHI using a robotic hand, further validating this approach. However, while Arata et al.'s robotic hand was limited to tapping the index finger, our system allows simultaneous and individual movement of all five fingers, potentially contributing to a more complex and robust illusion. Furthermore, although both studies employed the Botvinick and Cohen questionnaire, our study yielded higher scores, particularly for questions 3 ('I felt as if the rubber hand were my hand') and 4 ('It seemed as if I might have more than one right hand or arm'), compared to those reported by Arata et al. This difference may be attributed to the increased dexterity and realism afforded by our five-fingered robotic hand, which more closely mimics natural human hand movements and coordination.

The results of Experiment 2 indicate that the participants performed better in the memorization tasks with the robotic hand, compared to passively watching a video instruction, suggesting that physical interaction and embodiment effects contribute to cognitive engagement and learning efficiency. This aligns with existing research on active learning, which shows that physical involvement and sensory integration can improve memory and learning outcomes [30].

The study highlights the importance of agency in enhancing the sense of ownership and embodiment. Active control over the robotic hand not only deepened the illusion but also introduced a dynamic element to the participants' experience, merging sensory feedback with motor control to reinforce the integration of the robotic hand into the body schema. This suggests a promising avenue for rehabilitation technologies, emphasizing the importance of user control and interaction in designing effective assistive devices. Furthermore, the robotic hand provides an alternative to virtual reality for patients who are uncomfortable with or unable to use headsets [31].

VI. Future work

While the study provides valuable information on the effects of embodiment and technology on perception and learning, it is subject to limitations. The use of a predominantly student sample and the reliance on subjective measures of embodiment highlight the need for further research across diverse populations and with more objective measurement techniques.

Given the promising results in the implementation and performance of motor tasks, future work should prioritize clinical translation and patient-centered design. Although our study provided initial information using a brief experimental setup, conducting long-term impact studies with stroke patients in real-world rehabilitation settings is essential. These trials can provide valuable data on the effectiveness, safety, and user experience of the robotic hand in diverse patient populations, and would help evaluate the sustainability of the improvements achieved through robotic-assisted rehabilitation.

Researchers and therapists can track patient progress over long periods to assess the durability of motor recovery and functional improvements. Crucially, these clinical trials should incorporate detailed patient feedback on usability, comfort, and perceived benefits of the robotic hand. Understanding patient needs and preferences will be essential to tailor technology to individual rehabilitation goals.

Future studies should also explore the impact of personalized ergonomic design on user experience and adherence to rehabilitation protocols. Patient feedback can be used to inform design iterations, ensuring that the robotic hand is comfortable, intuitive to use, and effectively supports individual motor impairments.

Furthermore, involving therapists and medical experts can lead to a deeper understanding of the clinical implications and practical challenges associated with robotic-assisted rehabilitation. For example, therapists could provide insights into how the robotic hand can be best integrated into existing rehabilitation workflows and used to augment traditional therapy techniques.

To enhance connection and embodiment, both mechanical and cosmetic improvements are necessary. In particular, applying soft robotics could offer more natural movements and a lifelike appearance, potentially improving user experience and the illusion's effectiveness.

The insights gained from clinical trials and patient feedback can then be used to guide future technological development, ensuring that improvements are driven by real-world needs and lead to tangible benefits for stroke survivors. This iterative process of clinical evaluation, patient feedback, and technological refinement will be critical to realizing the full potential of robotic embodiment in stroke rehabilitation.

VII. Conclusion

During this study, a successful prototype robotic hand was built using inexpensive 3D-printed parts and appliances. The hand was able to mimic the opening and closing of fingers of a human hand with a latency of only 90ms, making it suitable to be used as a tool to research embodiment with active versus passive stimuli, as well as memory retrieval.

The results indicate that the active version of the robotic hand created a higher sense of embodiment than the passive version. Additionally, the robotic hand yields significantly higher accuracy in memorization tasks compared to video clip instructions.

In conclusion, the results of this study suggest that the robotic hand could be a valuable and useful tool for stroke patients during their rehabilitation process, adding a personalized experience with the help of active stimulation combined with low costs.

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